

PROJECT REPORT

E-Commerce Orders Data Analysis & Dashboard

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1. Executive Summary

This project involved the analysis of an e-commerce orders dataset to uncover key business insights. Using Excel, Power Query, MySQL, and Power BI, I performed data cleaning, exploratory data analysis, and created an interactive dashboard.

Key Results:

- Total Revenue: \$1.26 Million
- Total Orders: 1,000+
- Average Order Value: \$1,260
- Top Referral Source: Instagram
- Notable cart abandonment and order cancellation rates

2. Project Objectives

- Clean and transform raw e-commerce data
- Perform in-depth analysis using SQL
- Build an interactive Power BI dashboard
- Provide actionable business recommendations

3. Tools Used

Power Query; Data transformation

Power BI; Visualization & Dashboard

MySQL; Exploratory Data Analysis (EDA)

Excel; Initial exploration & cleaning

4. Data Cleaning & Preparation

- Standardized data formats (especially dates)

- Removed duplicates
- Created calculated columns where necessary
- Used Power Query for efficient transformations

5. Key SQL Analysis Performed

- Revenue analysis by product, referral source, and payment method
- Order status breakdown (Shipped, Delivered, Cancelled, Returned)
- Coupon code effectiveness
- Cart abandonment insights (Items in Cart vs Items Purchased)
- Top performing products and customers

6. Power BI Dashboard Highlights

The interactive dashboard shows:

- Overall KPIs
- Revenue by Payment Method
- Product Performance
- Cart Abandonment Analysis
- Revenue Trend Over Time
- Referral Source Performance
- Order Status Breakdown

7. Key Insights & Business Recommendations

1. Focus marketing efforts on Instagram (highest revenue driver).
2. Investigate and reduce high cancellation/return rates.
3. Improve checkout process to reduce cart abandonment.
4. Promote top-performing products more aggressively.
5. Optimize coupon strategy for better revenue impact.

8. Challenges Faced

- Working with messy raw data

- Converting Excel serial dates
- Balancing detailed analysis with clear visualizations

9. Conclusion

This project strengthened my skills in data cleaning, SQL querying, and business intelligence using Power BI. The insights generated can help the business improve revenue, reduce losses, and enhance customer experience.

Thank You!